IPE assignment

Ch-5 (Capacity planning)

Critical thinking – 1, 2 (Quiz থেকে)
Example 3 (book page 204)
Problem – 3, 4, 6 (book page 212)

Ch-5 (supplement)

Problem – 1, 2, 4, 6, 13 (book page 231)

Ch-17

Critical thinking – 1, 2 (book page 772)
Problem – 1, 4, 5 (book page 772)

Ch-18

Critical thinking – 1, 2 (book page 815)
Problem – 1, 2 (book page 816)

09.10.2025
IPE

Critical thinking (imp. for C.T.)

① # An organization is facing fluctuating customer demand and considering whether to expand its production capacity immediately or in several smaller phases. Analyze this situation using the principles of capacity planning and factors affecting capacity ~~decith~~ decisions. Which approach would you recommend and why?

⇒

② # Suppose a manufacturing firm has high design capacity but low effective capacity due to frequent machine breakdowns, poor scheduling, and low employee morale. Using understanding of the "Determinants of Effective capacity." analyze the root cause & suggest a comprehensive action plan to improve capacity utilization and efficiency.

Example - 03

Given,

Fixed Cost, $FC = \$6000$

Variable Cost, $VC = \$2$ per pie

Revenue, $R = \$7$ per pie

(a) How many pies must be sold in order to break even?

$$\Rightarrow Q_{BEP} = \frac{FC}{R - VC}$$

$$= \frac{6000}{7-2}$$

$$= \frac{6000}{5}$$

$$= 1200 \text{ pies/month}$$

(Ans)

(b) What would the profit (loss) be if 1,000 pies are made and sold in a month?

$\Rightarrow$ Profit, $P = Q(R - VC) - FC$

For $Q = 1000$,

$$P = 1000(7-2) - 6000$$

$$= -1000$$

$\therefore$ loss $= \$1000$

(Ans)

(c) How many pies must be sold to realize a profit of \$4,000 ?

$\Rightarrow P = Q(R - VC) - FC$

$\Rightarrow 4000 = Q(7-2) - 6000$

$\Rightarrow 5Q = 4000 + 6000$

$\Rightarrow Q = 10000/5$

$\therefore Q = 2000$ pies (Ans)

(d) If 2000 can be sold, and a profit target is \$5000, what price should be charged per pie?

$\Rightarrow$ Profit, $P = Q(R - VC) - FC$

$\Rightarrow 5000 = 2000(R-2) - 6000$

$\Rightarrow 2000(R-2) = 11000$

$\Rightarrow R - 2 = 11000/2000$

$\Rightarrow R = (11000/2000) + 2$

$\therefore R = \$7.50$

(Ans)

Problem - 03

Given,

Fixed Cost, FC = \$ 9200

Variable cost, VC = 70 cents
= \$ 0·7

Revenue, R = 90 cents
= \$0·9

(a) What volume per month is required in order to break even?

$$\Rightarrow Q_{BEP} = \frac{FC}{R-VC}$$

$$= \frac{9200}{0.9-0.7}$$

$$= \frac{9200}{0.2}$$

$$= 46000 \text{ units}$$

(Ans)

(b) What profit would be realized on a monthly volume of 61,000 units? 87,000 units?

⇒ We know,

Profit, $P = Q\,(R - VC) - FC$

For $Q = 61,000$,

$$P = 61000(0.9 - 0.7) - 9200$$
$$= 12200 - 9200$$
$$= 3000$$

$\therefore$ Profit \$3,000 for 61000 units.

For $Q = 87,000$,

$$P = 87,000(0.9 - 0.7) - 9200$$
$$= 8200$$

$\therefore$ Profit \$ 8200 for 87,000 units.

(Ans)

(c) What volume is needed to obtain a profit of \$ 16,000 per month?

$$\Rightarrow P = Q(R - VC) - FC$$
$$\Rightarrow 16000 = Q(0.9 - 0.7) - 9200$$
$$\Rightarrow Q(0.2) = 25200$$
$$\Rightarrow Q = 25200/0.2$$
$$\therefore Q = 126000 \text{ units}$$

(Ans)

(d) What volume is needed to provide a revenue of \$23,000 per month?

⇒ We know,

Total revenue, $TR = R \times Q$

$$\therefore Q = \frac{TR}{R}$$

$$= \frac{23000}{0.9}$$

$$= 25,556 \text{ units}$$

(Ans)

(e) Plot the total cost and total revenue lines.

Problem - 04

Given,

Two alternetives, A & B

For A,

Fixed cost, FC = \$40,000

Variable cost, VC = \$ 10

Revenue, R = \$ 15

For B,

Fixed Cost, FC = \$ 30,000

Variable Cost, VC = \$ 11

Revenue, R = \$15

(a) Determine each alternative's break-even point in units.

⇒ For A,

$$Q_{BEP} = \frac{FC}{R-VC}$$

$$= \frac{40000}{15-10}$$

$$= \frac{40000}{5}$$

$$= 8000 \text{ units}$$

(Ans)

For B,

$$Q_{BEP} = \frac{FC}{R-VC}$$

$$= \frac{30000}{15-11}$$

$$= \frac{30000}{4}$$

$$= 7500 \text{ units}$$

(Ans)

(b) At what volume of output would the two alternatives yield the same profit?

(b)

For A

$$P = Q(R-V) - FC$$

$$= Q(15-10) - 40000$$

$$= 5Q - 40000$$

For B

$$P = Q(15-11) - FC$$

$$= 4Q - 30000$$

$$\therefore 5Q - 40000 = 4Q - 30000$$

$$\Rightarrow 5Q - 4Q = 40000 - 30000$$

$$\therefore Q = 10000$$

(a)

(c)

For A

$P = Q(R-V) - FC$

$= 12000(15-10) - 40000$

$= \$20000$

For B

$P = 12000 \times 4 - 30000$

$= \$18000$

A would yeild the highest profit.

Problem 6

Solution:

$$17 + 18 + 15 = 50$$

Ans: 50 units/hrs

Supplement to Chapter 5

Decision Theory

Problem 1

Solution:

2

Expand (80).

6

Do nothing (50).

(c)

	Row Total	Row Average
Do nothing	110	55
Expand	100	50
Subcontract	110	55

~~Do noth~~ Do nothing (55) or Subcontract (55).

(d)

Alternatives	Regrets		
	Low	High	Worst
Do nothing	0	20	20
Expand	30	0	30
Subcontract	10	10	10

Subcontract (10).

Problem 2

Solution:

(a)

$$EMV_{\text{Do nothing}} = 50(0.3) + 60(.7)$$
$$= 57$$

$$EMV_{\text{Expand}} = 20(.3) + 80(.7)$$
$$= 62$$

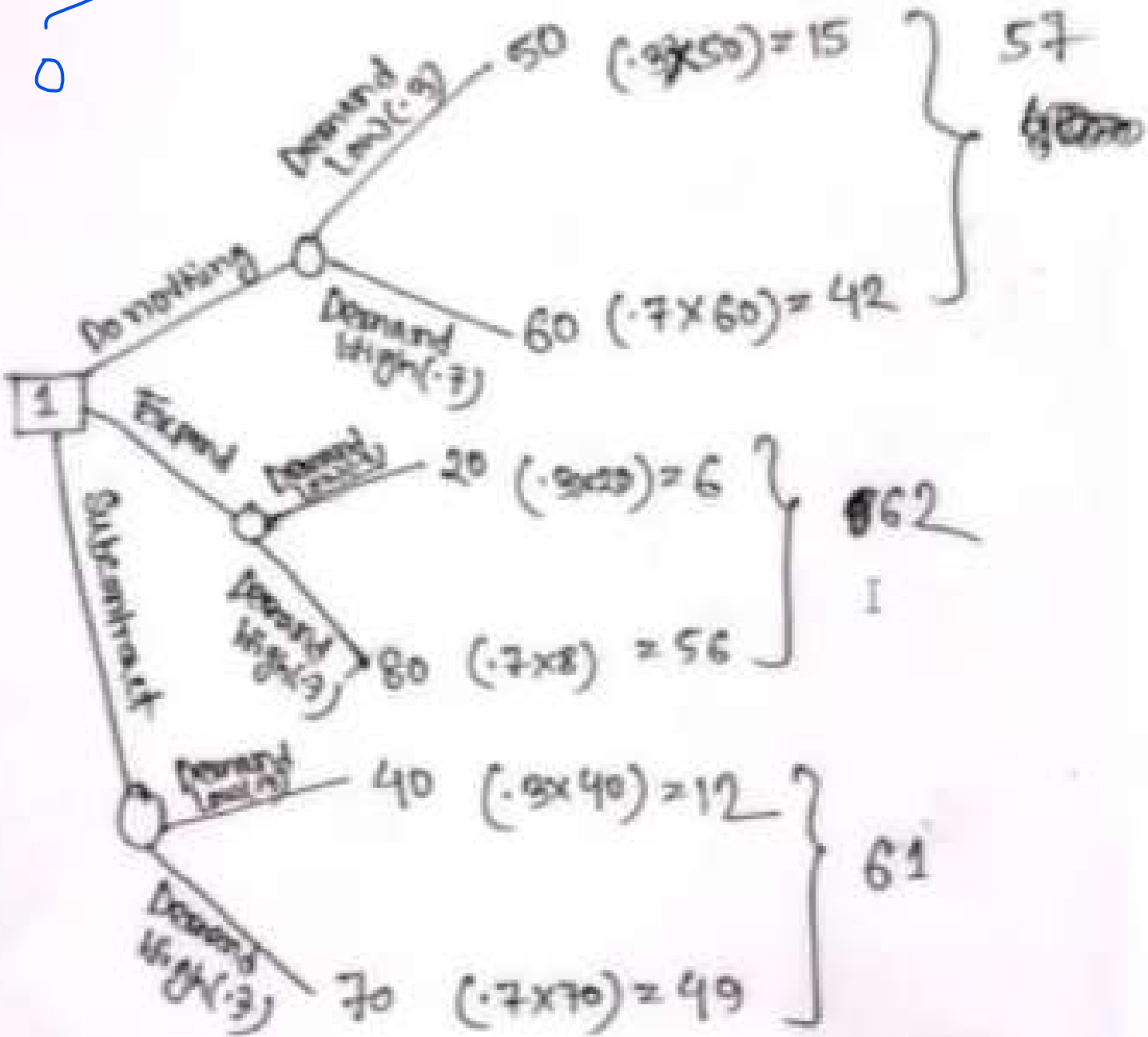
$$EMV_{\text{subcontract}} = 40(.3) + 70(.7)$$
$$= 61$$

Expand (B4) is the best.

(b)

! ?

Sure Na


1
Do nothing
Expand
Subcontract
Demand high (.3)
50 (.9x50) = 15
60 (.7x60) = 42
57
20 (.9x20) = 6
80 (.7x8) = 56
62
40 (.9x40) = 12
70 (.7x70) = 49
61

(c)

Expected payoff under certainty

$= (.3 \times 50) + (80 \times .7)$

$= 15 + 56 = 71$

$\therefore$ EVPI $= 71 - 62 = 9$.

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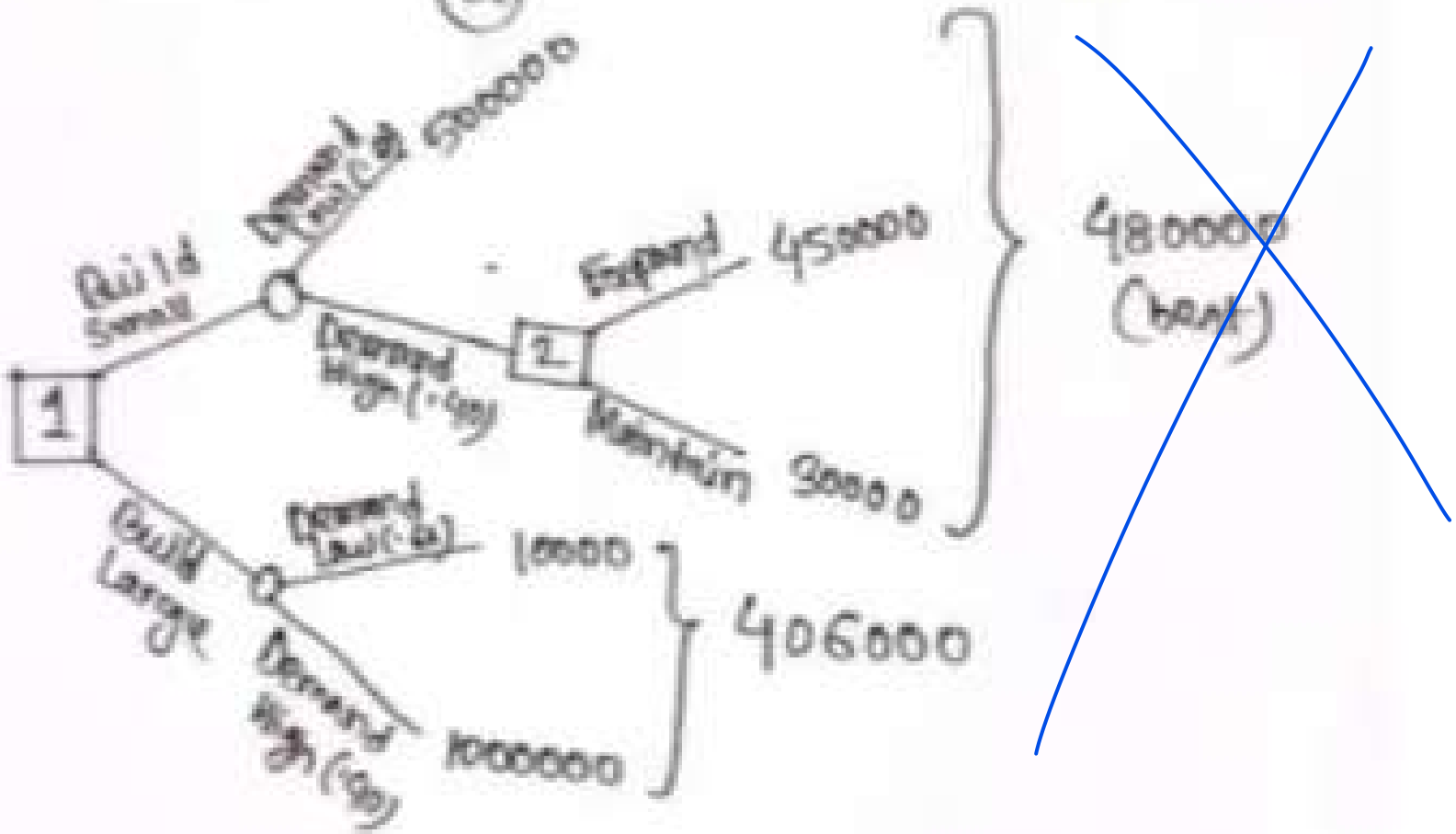
(8)

Problem 4

Solution:

???

(a)



(b)

Expected payoff under certainty

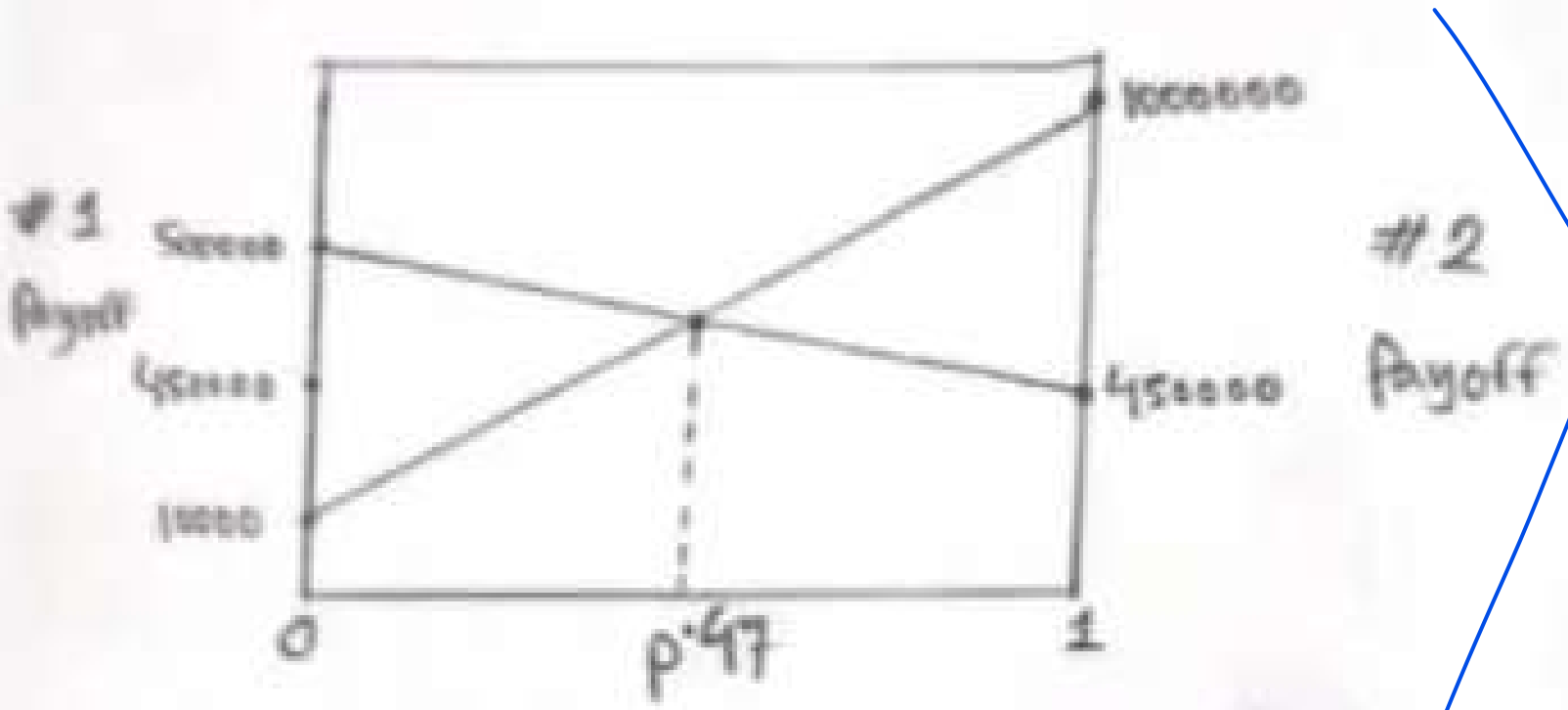
$= (.60)(500000) + (.40)(1000000)$

$= 700000$

$\therefore$ EVPI $= 700000 - 480000$

$= 220000$

This EVPI provides perfect information. On the basis of this information the managers will take decision.



Small facility

$500000 - 50000p$

Large facility

$$10000 + 990000\,p$$

$$\therefore\ 990000\,p + 50000\,p = 500000 - 10000$$

$$\Rightarrow\ p = \frac{490000}{1040000}$$

$$\therefore\ p = 0.47$$

P(Large) 0.47 to 1

P(Small) 0 to 0.47

Problem 6

Solution :

(a)

Relocate (5000000).

(b)

Renew (500000).

(c)

	Row Total	Row Average
Renew	900000	450000
Relocate	5010000	2505000*

Relocate (2505000).

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(d)

Options	Regrets		Worst	
	Value Admitted	Value Rejected		
Remain	4500000	0	4500000	
Relocate	0	390000	390000	Relocate (990000).

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(11)

Sure Na

Problem 13

Solution:

(a)

New staff (60).

(b)

~~Redis~~ Redesign collection (90).

(c)

(c)

Alternatives	Moderate	High	Very High	Worst
Redesign Staff	10	0	05	10
New Staff	0	0	30	30
Redesign Collection	20	10	0	20

Redesign staff (10).

(d)

	Row Total	Row Average
Redesign staff	195	65
New Staff	180	60
Redesign collection	180	60

Redesign Staff (65).

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Critical Thinking 1

Soution :

It is quite true that project management techniques have been used successfully in the given sectors. However it doen't mean that its techniques are not as much as useful in operation management. It can be even more useful in operation management of any business such as designing new products or services, information systems, advertising, reengineering process and so on.

The project management techniques enable an organization to focus attention and concentrate efforts on accomplishing a set of objectives within given time and budget frame. These techniques will have strategic importance for organizations. For instance good project management can be instrumental in successfully

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implementing an enterprise resource planning system are converting a traditional operation to lean operation.

The major uses of ~~opera~~ project management techniques in any business are described below:

1. Planning and Control : Project management techniques (PMT) are widely used in planning and controlling the available resources in any business. The business performance need to be monitored and progress need to be measure and for this, project management techniques could be used.

2. Time Management: Time management is the process of making sure the project ~~Shedule~~ schedule doesnot slip and it is on time. For managing time, PMT are used so that timely decisions can be made and demand can be fulfilled according to customers needs.

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3. Cost Management: In order to reduce the costs, PMT could be used so that no extra costs take place during the production or operation of business activities.

4. Quality Management: Quality products or services need to be produced and delivered them to customers. However, it would not be possible without the use of project management techniques such as statistical analysis and statistical process control for quality measure.

5. Performance Management: Performance management is one of the key tasks for any business organization. With the help of project management techniques, performance and progress can be measured and instant actions will be taken if necessary.

(5)

~~Critical Thing 2~~

Critical Thinking 2

Solution :

Ethical Principle 1 : Labor laws shouldn't be violated.

Example: A project is running late and the project manager pushes the employees to work hours without compensating adequately. The employees fear to lose their job if they don't agree to what project manager says.

Ethical Principle 2: No consideration for society and the environment.

Example: The project manager hired the sub-contractor who issuing the polluting chemicals for the project work. This chemicals being discharged to nearby water body after use which may danger the lives of communities living around the project site.

Ethical Principle 3: Indulging in unethical practices like bribery.

Example: Project Managers has accepted an expensive gift from a sub-contractors and now he is obliged to hire this sub-contractors the project even though the subcontractors doesn't meet all the criteria.

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(b)

Problem 1

Solution:

(a)

Path	Expected Duration
1-2-4-7-10-12	4+5+2+3 = 23
1-2-5-8-10-12	4+8+7+2+3 = 24
* 1-3-6-9-11-12	10+6+4+5+6 = 31*

(b)

Path	Expected Duration
1-2-4-6-8-9	5+18+3+4+9+2 = 41
1-2-5-6-8-9	5+18+10+4+9+2 = 48
* 1-2-5-7-8-9	5+18+10+11+9+2 = 55*
1-3-7-8-9	5+19+11+9+2 = 40

(c)

1-2-5-12-16	10+14+13+7 = 44
1-3-6-13-16	14+15+6+4 = 39
1-3-7-14-16	14+11+13+3 = 41
1-4-8-9-10-11-15-16	9+8+1+0+7+6+10 = 35
1-4-8-10-11-15-16	3+8+4+7+6+10 = 38

(d)

A-D	7+8=15
B-E	3+4=7
C-F-E	5+7+4=16
C-F-G	5+7+6 = 18

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Problem 4

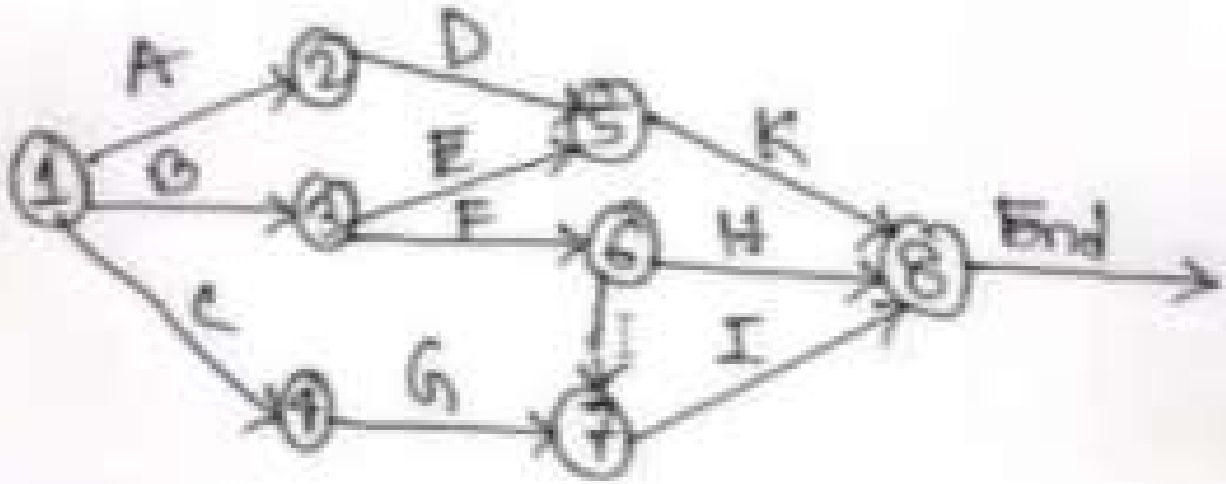
Solution:

(a)

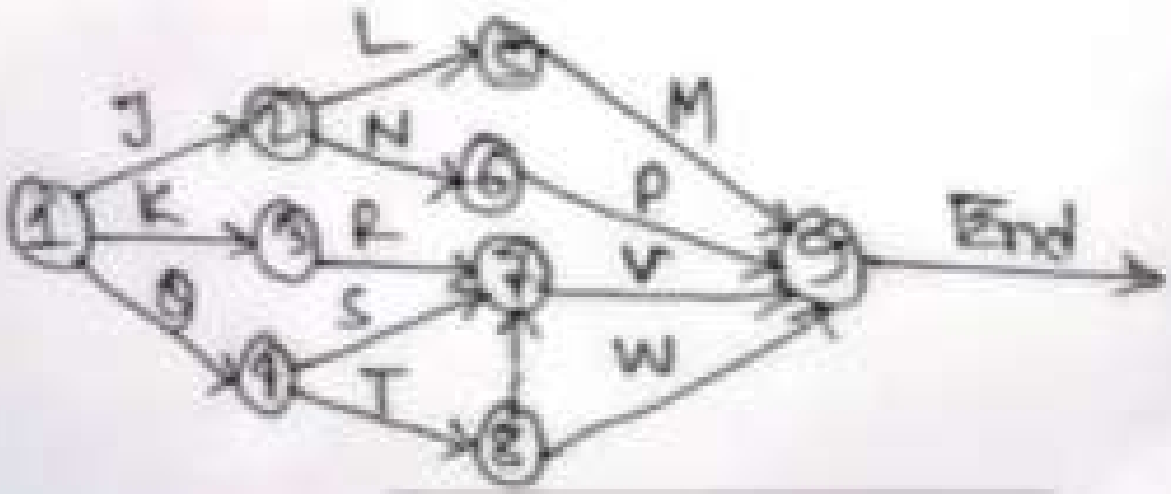
Activity	Immediate Predecessors
A	—
B	—
C	A
E	C
F	A, B
End	E, F

(b)

Case 1: Activity on Arrow Diagram



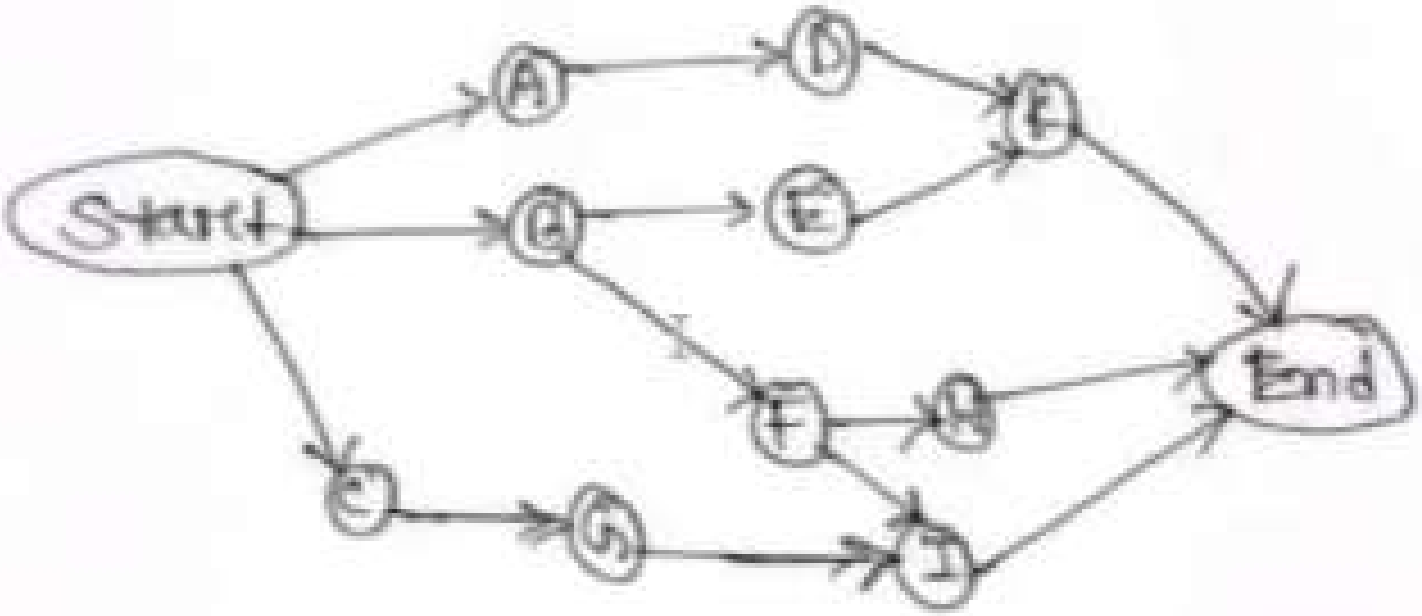
Case 2: Activity on Arrow Diagram



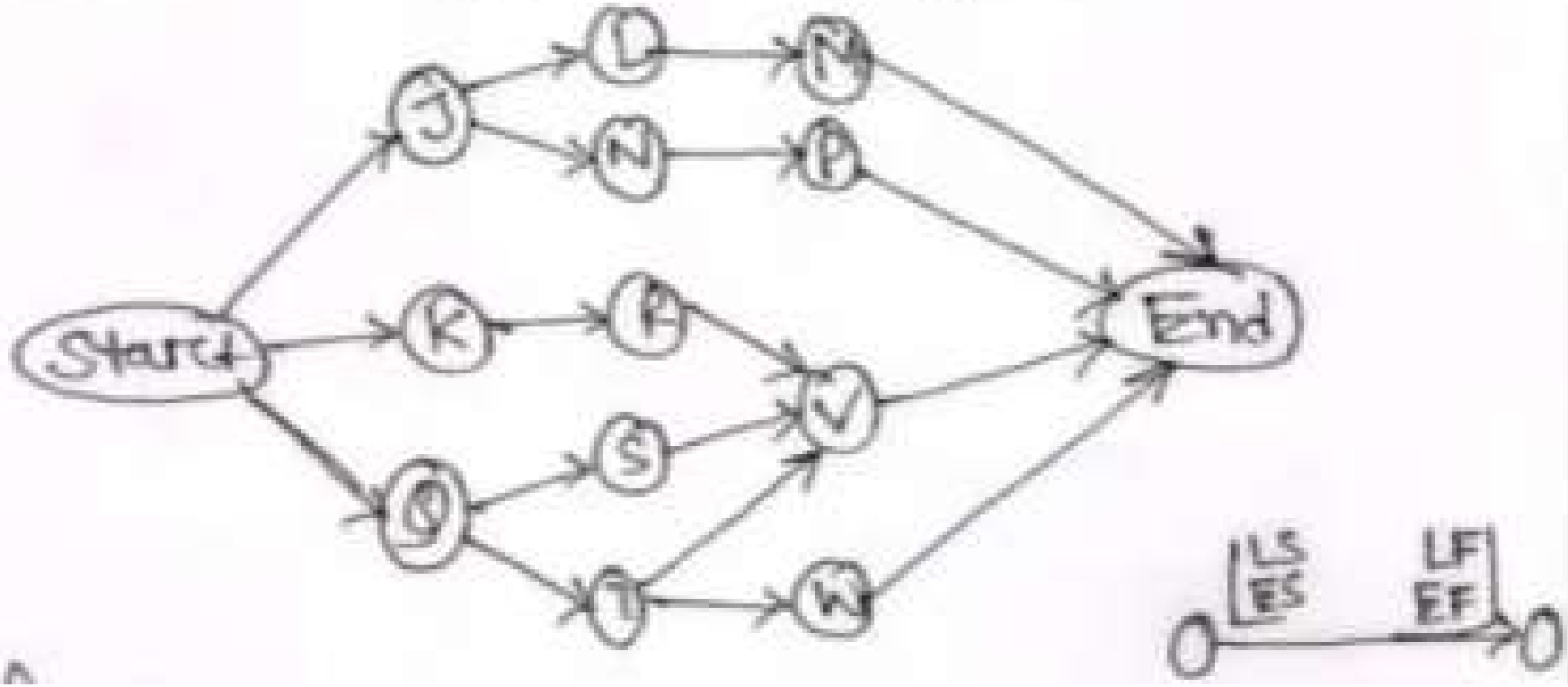
(8)

(c)

Case 1: Activity on Node Diagram

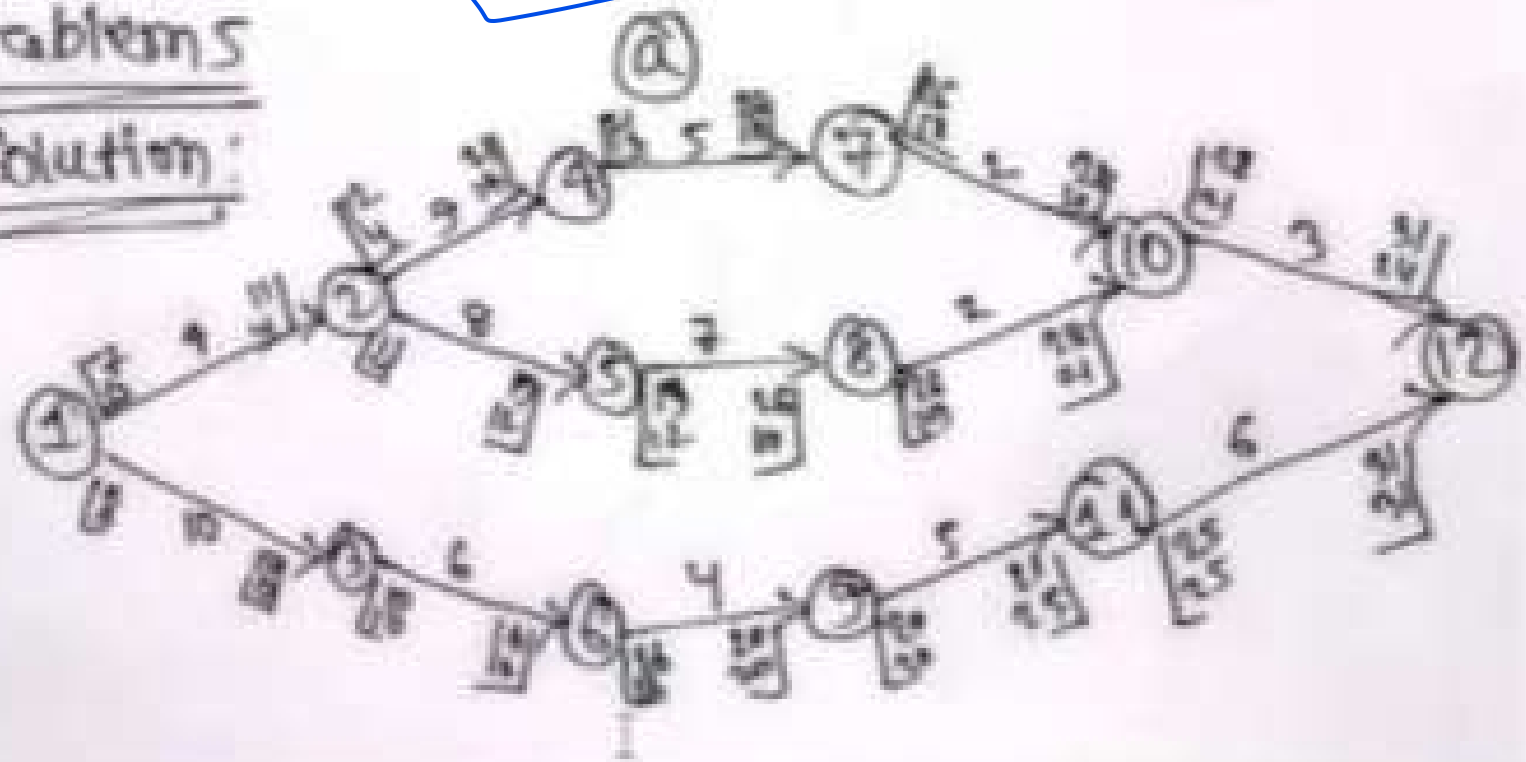


Case 2: Activity on Node Diagram



Problem 5

Solution:



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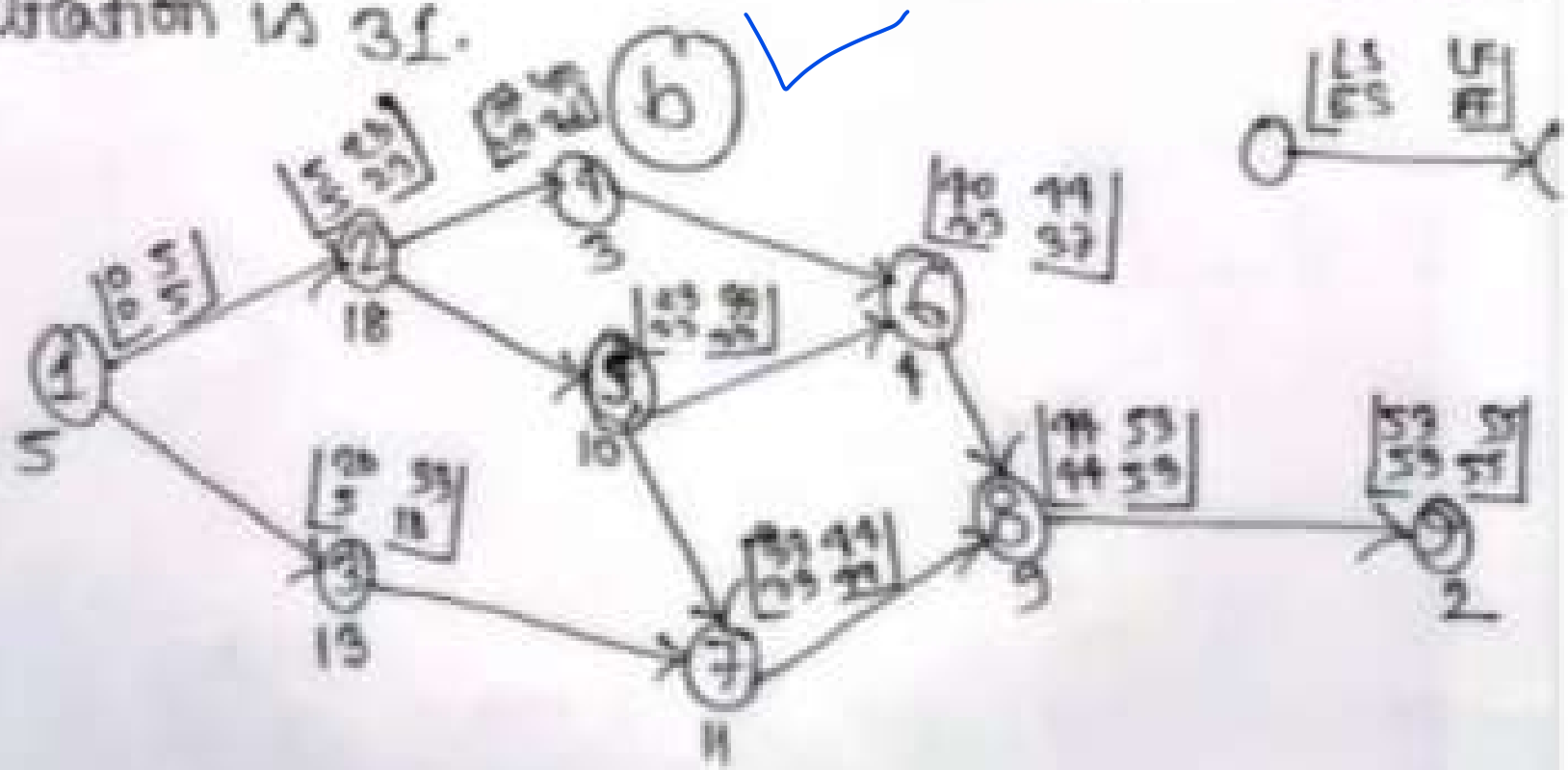
Slack = LS − ES or LF − EF

Activity	ES	EF	LS	LF	Slack
1-2	0	4	7	11	7
1-3	0	10	0	10	0
2-4	4	13	12	21	8
2-5	4	12	11	19	7
3-6	10	16	10	16	0
4-7	13	18	21	26	8
5-8	12	19	19	26	7
6-9	16	20	16	20	0
7-10	18	20	26	28	8
8-10	19	21	26	28	7
9-11	20	25	20	25	0
10-12	21	24	28	31	7
11-12	25	31	25	31	0

Critical path is 1-3-6-9-11-12. Expected project duration is 31.



Slack = LS - ES or LF - EF

Activity	ES	EF	LS	LF	Slack
1	0	5	0	5	0
2	5	23	5	23	0
3	5	18	20	33	15
4	23	26	37	40	14
5	23	33	23	33	0
6	33	37	40	44	7
7	33	44	33	44	0
8	44	53	44	53	0
9	53	55	53	55	0

Critical path is 1-2-5-7-8-9. Expected Project durcation is 55.

Chapter 18

Management of Waiting Lines

Critical Thinking 1

Solution:

Psychology of waiting plays an important part in determining the queuing model so as to minimize the total waiting and service cost. Since it is known that the psychology of waiting affects the ultimate customer satisfaction, many organizations have taken various steps to alleviate the negative impact of the waiting.

The following approaches are derived regarding the psychology of waiting:

1) If a person is doing something then the wait time seems shorter. Occupied time always seems shorter than unoccupied time.

2) If a person is aware about the waiting time then it is easier for the individual to manage time.

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3) Waiting alone always seems longer than if there is a group

4) Anxiety creates the waiting ~~time~~ time longer than it may seem.

5) Unexplained and unfair waits are always longer than explained and equitable waits

6) More the values of the service to the customer, longer he or she would be willing to wait.

To improve upon these psychologies with respect to waiting system, better queuing system is created. The service and arrival rate can be taken into consideration, to improve upon the waiting and service levels to customers.

Thus, greater benefits can be derived by organizations who acknowledge and use the psychology of waiting to their advantage.

(73)

Critical Thinking 2

Solution:

(a)

We know that as utilization increases, waiting times increas exponentially. Conversely when utilization decreases (as would be the case when service rate doubles), we would expect a disproportionate decrease in the waiting time i.e. greater than 50%.

(b)

With an arrival rate of 8 customers per hour and a service rate of 10 customers per hour:

$$L_q = \frac{(8)^2}{10(10-8)} = 3.2 \text{ customers/hour}$$

With an arrival rate of 8 customers per hour and a service rate of 20 customers per hour:

$$L_q = \frac{8^2}{20(20-8)} = 0.267 \text{ customers/hour}$$

The decrease in the average number in line $= 3.2$.

(c)

Doubling the service rate is definitely a good solution but that can be done only at cost of adding another server. Since the requirement demands, we are going for doubling the service rate. When customers are to pay more, they expect service to be of greater quality. As stated increasing in the service rate would actually cut the average number waiting in line almost in half.

As a manager, when there is a demand, I would opt to utilize the demand and increase the service rate, at the same time give good quality service and would work out ways to completely utilize the server space.

I would definitely suggest implementing another server.

③

$$W_L = \frac{L_q}{\lambda} = \frac{1.929}{18} = 0.107 \text{ hours.}$$

④

$$W_S = \frac{L_S}{\lambda} = \frac{3.429}{18} = 0.1905 \text{ hours}$$

Problem 1

Solution

(a)

(1)

$$\rho = \frac{\lambda}{M\mu} = \frac{3}{1\times 5} = 0.60$$

(2)

$$L_q = \frac{\lambda^2}{\mu(\mu-\lambda)} = \frac{3^2}{5(5-3)} = 0.9 \text{ customers}$$

(3)

$$W_q = \frac{L_q}{\lambda} = \frac{0.9}{3} = 0.3 \text{ hours}$$

(b)

(1)

$$\lambda = \frac{3}{8} = 0.375$$

$$\mu = 0.5$$

$$L_q = \frac{(0.375)^2}{0.5(0.5-0.375)} = \frac{0.140625}{0.0625} = 2.25 \text{ customers}$$

(2)

$$\rho = \frac{0.375}{1(0.5)} = 0.75$$

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(3)

$W_q = \lambda L_q = 0.375 \times 2.25 = 0.84375$ hours

$W_s = W_q + \frac{1}{\mu} = 0.84375 + \frac{1}{0.5} = 2.84375$ hours

$\therefore W_s - W_q = 2.84375 - 0.84375$

$= 2$ hours

(4)

$P_0 = 1 - \frac{\lambda}{\mu} = 1 - \frac{0.375}{0.5} = 0.25$

$P_1 = P_0 \left(\frac{\lambda}{\mu}\right)^1 = 0.25 \times \frac{0.375}{0.5} = 0.1875$

$\therefore$ Probability of two or more customers in the system

$= 1 - (0.25 + 0.1875) = 0.5625$

(c)

(1)

$\lambda = 18$ customers per hour

$\mu = 12$ " " "

$\rho = \frac{18}{2 \times 12} = 0.75$

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(2)

Let,

$L_q = 1.929$ customers

$\therefore L_s = L_q + \rho$

$= 1.929 + \frac{\lambda}{\mu}$

$= 1.929 + \frac{18}{12}$

$= 3.429$ customers

(3)

$W_q = \frac{L_q}{\lambda} = \frac{1.929}{18} = 0.107$ hours.

(4)

$W_s = \frac{L_s}{\lambda} = \frac{3.429}{18} = 0.1905$ hour

Problem 2

Solution:

(a)

$\mu = \frac{1}{30} \times 60 \times 60 = 120$ customers/hour

$\lambda = 80$ customers per hour

$L_q = \frac{\lambda^2}{2\mu(\mu-\lambda)} = \frac{80^2}{2 \times 120(120-80)} = 0.67$ customer

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(28)

(b)

$$W_s = W_q + \frac{1}{\mu}$$

$$= \frac{L_q}{\lambda} + \frac{1}{\mu}$$

$$= \frac{0.67}{80} + \frac{1}{120}$$

$$= 1.0025 \text{ min}$$

$$= 1 \text{ min}$$

(c)

$$L_s = L_q + \rho$$

$$= 0.67 + \frac{80}{120}$$

$$= 1.34 \text{ customers}$$

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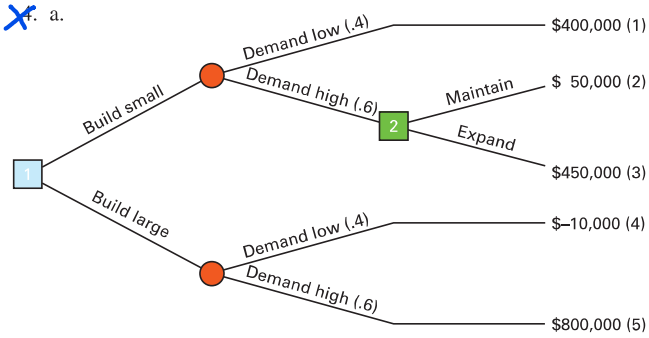
Chapter 5: Strategic Capacity Planning for Products and Services

1. a. Utilization = 70%.
   Efficiency = 87.5%.
   b. Utilization = 67%.
   Efficiency = 80%.
2. 20 jobs per week.
3. a. 46,000 units.
   b. (1) $3,000. (2) $8,200.
   c. 126,000 units. d. 25,556 units.
4. a. A: 8,000 units. b. 10,000 units.
   B: 7,500 units.
   c. A: $20,000.
   B: $18,000.
5. a. 39,683 units. b. $1.71 (rounded up).
6. a. A: $82.
   B: $92.
   C: $100.
   c. A: 0 to less than 178.
   B: Never.
   C: 178 + .
   d. 1/3 day, 2/3 evening.
7. Vendor best for $Q < 63{,}333$. For larger quantities, produce in-house at $4 per unit.
8. a. Vendor B best for 10,000 and 20,000.
9. 3 cells.
11. a. Buy 2 Bs. b. Buy 2 Bs.
12. a. one: $Q = 80$. two: $Q = 152$.
14. a. 11 units/hr.
16. 50 units/hr.
17. a. 15 units/hr. b. 23 units/hr.
18. 8 years.

Chapter 5 Supplement: Decision Theory

1. a. Expand (80).
   b. Do nothing (50).
   c. Indifferent between do nothing and subcontract (55).
   d. Subcontract (10).
2. a. Expand (62). c. $9 (000).
3. Do nothing: $P(\text{high}) < .50$.
   Subcontract: $.50 < P(\text{high}) < .67$.
   Expand: $P(\text{high}) > .67$.
4. a.



   b. $164,000.
   c. Large 0 to .46.
Small .46 to 1.00.

5. Subcontract: $1.23.
   Expand: $1.57.
   Build: $1.35.
6. a. Relocate.
   b. Renew.
   c. Relocate.
   d. Relocate.
7. a. Renew.
   b. EVPI = $1,575,000.
   c. Yes.
9. a. Build large: $53.6 million.
   b. Build small: $42 million.
   c. $12.4.
   d. Build small for $P(\text{high}) < .721$.
   Build large for $P(\text{high}) > .721$.
10. Buy two ($113.5).
11. A: 49.
12. b. maximin: small.
    maximax: large.
    Laplace: large.
    minimax regret: large.
13. a. New staff.
    b. Redesign.
    c. New staff.
    d. New staff or redesign.
16. b. Alternative C.
    c. $P(2) > .625$.
    d. $P(1) < .375$.
17. b. Alternative B.
    c. $P(2) < .444$.
    d. $P(1) > .556$.

Chapter 6: Process Selection and Facility Layout

1. a. Minimum is 2.4 minutes, maximum is 18 minutes.
   b. 25 units to 187.5 units. c. Eight.
   d. 3.6 minutes. e. (1) 50 units.
   (2) 30 units.
2. a.

Station	Tasks	Time
1	a	1.4
2	b, e	1.3
3	d, c, f	1.8
4	g, h	1.5

3. a.

Station	Tasks	Time
1	f, a, g	14
2	d, b, c	13
3	e, h	13
4	i	5

4. a. (3) 11.54%. (4) 323 copiers per day.
   b. (1) 2.3 minutes. (3) 182.6 copiers per day.
   (4) 91.3 copiers units per day.
5. b. 2 minutes.
   c. Three stations.
6. c. (1) 11.1%. (2) 11.1%.
7. b. CT = .84 min. or 50.4 sec.
   c. $n = 3.83$ (round to 4) stations.

13. G-A-E-D-B-C-F (or G-E-A-D-B-C-F).
14. a., b. Grinding flow time is 93 hours. Total time is 37 hours.
    c. Grinding flow time is 107 hours. Total time is 35 hours.
15. a.

	FCFS	SPT	EDD	CR
Av. flow time.	15.25	11.12	15.60	15.60
Av. job tardiness.	1.7	2.33	0.55	0.55
Av. no. of jobs.	3.91	2.85	4.00	4.00

    c. a-c-b-e-d-f.
16. b-c-e-a-d.
19. A-B-C.
20. C-B-A.
21. B-C-D-A.

Chapter 17: Project Management

1. a. 1-3-6-9-11-12: 31.
   b. 1-2-5-7-8-9: 55.
   c. 1-2-5-12-16: 44.
4. a.

Activity	Immed. Pred.
A	–
B	–
C	A
F	A,B
E	C

5. a. Summary:

Activity	ES	EF	LF	LS	Slack
1-2.	0	4	11	7	7
2-4.	4	13	21	12	8
4-7.	13	18	26	21	8
7-10.	18	20	28	26	8
10-12.	21	24	31	28	7
2-5.	4	12	19	11	7
5-8.	12	19	26	19	7
8-10.	19	21	28	26	7
1-3.	0	10	10	0	0
3-6.	10	16	16	10	0
6-9.	16	20	20	16	0
9-11.	20	25	25	20	0
11-12.	25	31	31	25	0

   b. Summary:

Activity	ES	EF	LF	LS	Slack
1.	0	5	5	0	0
2.	5	23	23	5	0
3.	5	18	33	20	15
4.	23	26	40	37	14
5.	23	33	33	23	0
6.	33	37	44	40	7
7.	33	44	44	33	0
8.	44	53	53	44	0
9.	53	55	55	53	0

6. 30 weeks.
7. b 24 days: .9686; 21 days: .2350.
   c. 24 days: .9328; 21 days: .0186.
   d. Crash activities F, C, and G one day each.
8. a. .6881.
   b. .3978.
   c. .0203.
9. .3479.
10. b. .52.
11. b. .030.
    c. .2085.
12.

Path	Mean	Standard Deviation
a-d-e-h.	24.34	1.354
a-f-g.	15.50	1.258
b-i-j-k.	14.83	1.014
c-m-n-o.	26.17	1.658

    27 weeks: .6742; 26 weeks: .4099.
14. Crash schedule (1 week each): C, C, F, F, E, P.
15. a. Crash four weeks:
    (1) 7-11, (2) 1-2, (3) 7-11 and 6-10, (4) 11-13 and 4-6.
17. a. 18.5 days.
    b. .67.

Chapter 18: Management of Waiting Lines

1. a. 1) .60
      2) .90 customer
      3) .30 hr.
   b. 1) 2.25 customers.
      2) .75.
      3) Two hours.
      4) .5625.
   c. 1) .75
      2) 3.429 customers
      3) .107 hr.
2. a. 0.67 customer.
   b. One minute.
   c. 1.33 customers.
3. a. 6 minutes.
   b. 0.25.
   c. 2.25 customers.
4. a. Morning: 0.375 minute; .45.
      Afternoon: 0.678 minute; .54.
      Evening: 0.635 minute; .44.
   b. M: 4; A: 8; E: 5.
5. a. .45.
   b. .229.
   c. .509 hour.
   d. .28.
6. a. 4.444 trucks.
   b. 6.67 minutes.
   c. .711.
   d. 6 minutes.
   e. The system would be overloaded.
   f. 13.186.
7. a. One dock.
   b. No.